Service Request Workflow Performance Analysis

1. Executive Summary

This project analysed service request workflow data to assess operational performance, identify delays in request resolution, and evaluate the reliability of key completion metrics. The analysis focused on data cleaning, validation of date fields, recalculation of completion time, descriptive workflow analysis, and visual examination of delay patterns across service request categories.

The findings suggest that the original completion-time field required validation, as date inconsistencies and missing completion records affected metric reliability. After cleaning and recalculating completion time, the analysis showed that request resolution times were not evenly distributed, with some requests taking substantially longer than others. Differences across service categories also indicated that delays were concentrated in specific parts of the workflow rather than being uniformly spread across all requests.

Overall, the project highlights the importance of using validated process metrics, monitoring categories with persistent delays, and strengthening data entry controls to improve operational visibility and workflow management.

2. Business Context

Service request workflows are a common part of operational service delivery. Their effectiveness depends not only on how quickly requests are resolved, but also on whether the underlying data is accurate enough to support monitoring and decision-making.

In this case, the project was designed to examine service request data from an operational perspective. The aim was to understand whether reported completion metrics were reliable, where delays were occurring, and what insights could support process improvement.

3. Project Objective

The objective of this project was to assess workflow performance using service request data by:

- identifying and addressing data quality issues that affected completion-time reporting
- recalculating completion time using date fields to improve metric reliability
- analysing request resolution patterns to detect delays, bottlenecks, and long-running cases
- generating recommendations to support better workflow monitoring and service delivery

4. Data Overview and Preparation

The dataset consisted of service request records, including variables such as request category, request and completion dates, location information, and recorded completion time.

Initial review of the data identified several quality issues that affected the reliability of workflow analysis, including:

- missing completion dates
- duplicate records
- inconsistent date fields
- potential mismatch between recorded completion time and date-based calculation
- some missing suburb information

To improve analytical reliability, the data was cleaned and validated before analysis. This included:

- converting date fields into appropriate formats
- checking missing values and duplicate records
- removing or filtering records with invalid completion logic
- recalculating days_to_complete from date fields
- retaining a cleaned dataset for workflow performance analysis

This preparation stage was necessary because any analysis based on inaccurate completion-time measures would weaken the validity of the operational findings.

5. Analysis and Findings

5.1 Completion Time Reliability

The project found that the original completion-time field could not be assumed fully reliable without validation. Recalculating completion time from request and completion dates provided a more consistent basis for analysis.

This is an important operational point: if workflow metrics are not measured consistently, performance monitoring may misrepresent where delays actually occur.

5.2 Resolution Time Distribution

The analysis showed that request resolution times were not evenly distributed. Instead, the distribution appeared to contain a longer tail, meaning that while many requests were completed in a shorter period, a smaller subset remained unresolved for much longer.

This suggests that average completion time alone may not be sufficient for workflow monitoring, as extreme delays can be hidden inside aggregate metrics.

5.3 Category-Level Differences

Resolution time varied across service request categories. Some categories showed longer completion times than others, indicating that delays were likely concentrated in specific workflow segments rather than spread evenly across the entire process.

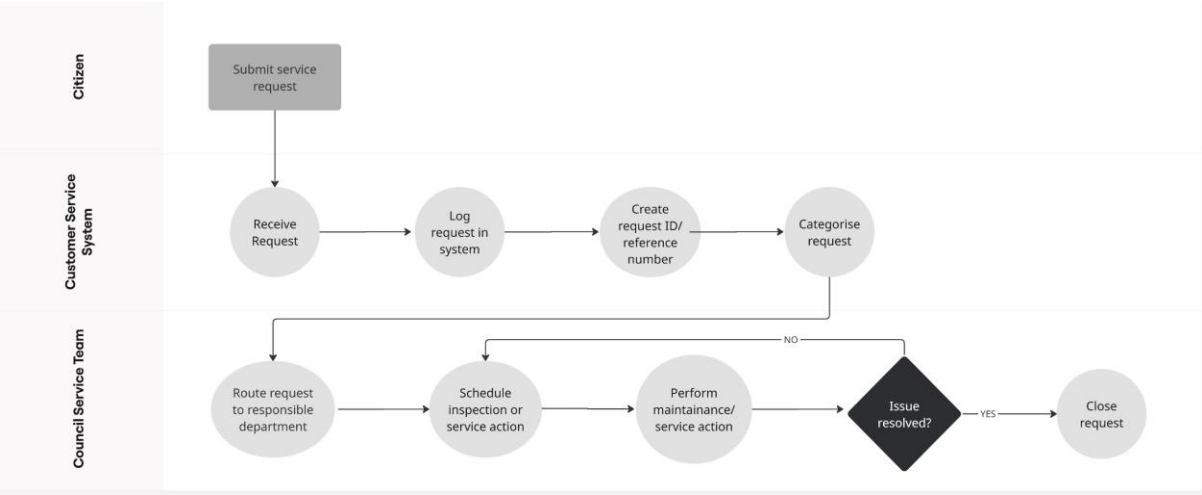
This type of pattern is operationally significant because it suggests that targeted intervention may be more effective than broad, organisation-wide process changes.

5.4 Long-Running Requests

The analysis also highlighted the presence of long-running requests that may have contributed disproportionately to workflow inefficiency. These requests are especially relevant for monitoring because they may represent backlog build-up, exceptions, or categories requiring escalation.

6. As-Is Process Snapshot

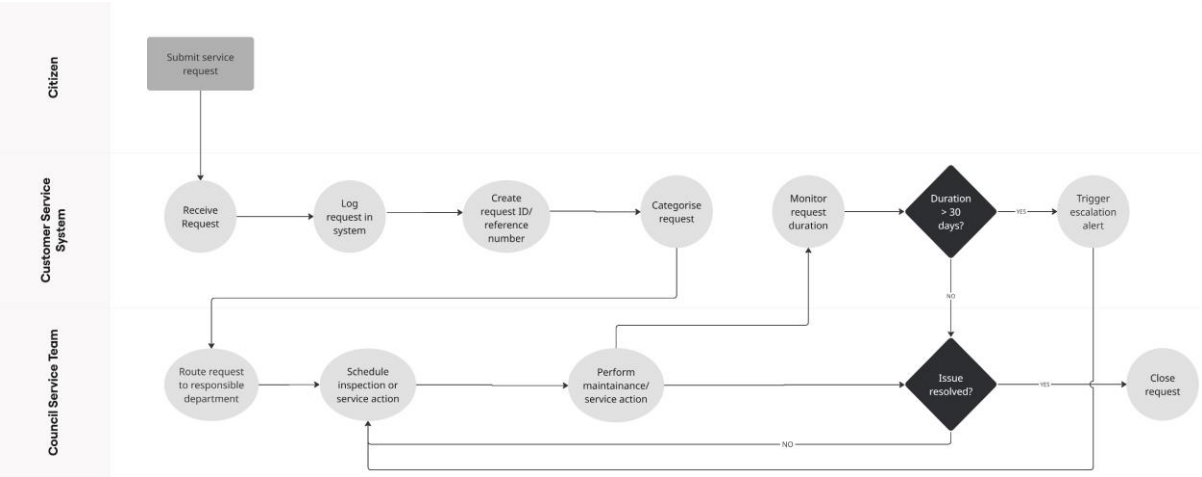
Current state observations (As-Is):



- Completion-time reporting required validation before it could be trusted for analysis.
- Workflow performance was uneven, with some requests taking significantly longer to resolve.
- Delays appeared more concentrated in certain request categories.
- Long-running requests may have reduced overall workflow efficiency.
- Data quality issues reduced the reliability of operational monitoring.

7. To-Be Process Direction

Desired future state (To-Be):



- Completion metrics should be derived from validated date logic rather than relying on potentially inconsistent reported fields.
- Workflow monitoring should include overdue or long-running request alerts, not only average completion time.
- Categories with persistent delays should be prioritised for process review.
- Data entry and completion-date validation should be strengthened to improve reporting quality.
- Operational dashboards should focus on delay distribution, category-level resolution time, and backlog risk indicators.

8. Recommendations

Based on the analysis, the following recommendations are appropriate:

8.1 Use Validated Completion Metrics

Operational monitoring should prioritise recalculated completion time based on date fields, especially where original recorded metrics are inconsistent.

8.2 Monitor Long-Running Requests Separately

Requests with unusually long completion times should be tracked separately from standard cases, as they may indicate bottlenecks, backlog concentration, or process exceptions.

8.3 Prioritise High-Delay Categories

Categories with consistently slower completion times should be reviewed first to determine whether delays reflect resource constraints, process complexity, or workflow design issues.

8.4 Improve Data Entry Controls

Validation checks for completion dates and related fields should be improved so that operational reporting is based on more reliable data.

8.5 Support Monitoring Through Dashboarding

A dashboard could be developed to track request volume, median completion time, overdue cases, and category-level delay patterns on a regular basis.

9. Limitations

This project has several limitations:

- some records contained missing values that reduced completeness
- the dataset captured workflow outcomes, but not the deeper operational reasons behind delays
- missing suburb or contextual fields limited some segmentation opportunities
- the analysis focused on historical descriptive patterns rather than causal explanation

As a result, the project is best interpreted as a workflow performance and monitoring analysis rather than a causal study of why delays occurred.

10. Conclusion

This project demonstrated how service request data can be used to assess workflow performance and identify operational bottlenecks. A key contribution of the analysis was validating and recalculating completion-time metrics before drawing conclusions, which improved the reliability of the results.

The findings indicate that service request delays were not uniform across the workflow and that a subset of requests and categories contributed more strongly to slower performance. From a business perspective, the project shows the value of combining data quality checks with operational analysis to improve performance monitoring, prioritisation, and process visibility.